

Quantum Mechanics – Homework 1 Solutions

Lectures 1–3

Instructor version. Numerical results are reported to three significant figures.

Problem 1. Measurements and Superposition (20 points)

- (a) The first color box prepares a definite White result. By repeatability, an immediate repetition of the same measurement gives

$$P(B) = 0, \quad P(W) = 1.$$

- (b) Because color and hardness are uncorrelated,

$$P(H | W) = P(S | W) = \frac{1}{2}.$$

Selecting Soft prepares a definite hardness state, which gives a 50–50 distribution when color is measured:

$$P(B | S) = P(W | S) = \frac{1}{2}.$$

Of the 1200 incident White electrons, 600 are expected at the Soft output and proceed to the last box. Therefore the expected final counts are

$$N_B = 600 \left(\frac{1}{2} \right) = 300, \quad N_W = 600 \left(\frac{1}{2} \right) = 300.$$

The other 600 electrons leave through the unselected Hard output.

- (c) With both paths open and coherently recombined, the apparatus restores the initial color result:

$$P(B) = 0, \quad P(W) = 1.$$

- (d) The Soft path contains half of the incident beam, so 600 electrons are absorbed. The remaining 600 electrons travel along the Hard path. A definite Hard state produces equal color probabilities, so

$$N_{\text{absorbed}} = 600, \quad N_B = 300, \quad N_W = 300.$$

Relative to the incident beam, the percentages are 50% absorbed, 25% Black, and 25% White.

- (e) The classical mixture predicts

$$\begin{aligned} P(B) &= P(B | H)P(H) + P(B | S)P(S) \\ &= \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{1}{2}. \end{aligned}$$

This disagrees with the observed value $P(B) = 0$ when both paths are open. The two alternatives therefore cannot be treated as an unknown classical choice of one path. Both path contributions must remain in the quantum description and combine as a superposition. This does not mean that the electron splits into two pieces; every final detection is still one whole electron.

Problem 2. Blackbody Radiation (20 points)

- (a) Radiation entering a small hole undergoes many reflections inside the cavity and is very unlikely to escape, so the hole behaves nearly as a perfect absorber. In thermal equilibrium the radiation leaving the hole depends mainly on the cavity temperature rather than on the material of the walls. When T increases,
- the total emitted power increases, and
 - the spectral peak moves to higher frequency (shorter wavelength).
- (b) Define $x = h\nu/(k_B T)$. When $x \ll 1$, $e^x - 1 \simeq x = h\nu/(k_B T)$. Substitution into Planck's formula gives

$$\begin{aligned} u_P(\nu, T) &\simeq \frac{8\pi h\nu^3}{c^3} \frac{1}{h\nu/(k_B T)} \\ &= \frac{8\pi\nu^2}{c^3} k_B T = \boxed{u_{RJ}(\nu, T)}. \end{aligned}$$

- (c) At fixed T , the Rayleigh–Jeans law grows without bound:

$$u_{RJ}(\nu, T) \propto \nu^2 \quad (\nu \rightarrow \infty).$$

Consequently, its predicted total energy $\int_0^\infty u_{RJ}(\nu, T) d\nu$ diverges. This is the ultra-violet catastrophe. For $h\nu \gg k_B T$, Planck's law instead has

$$e^{h\nu/(k_B T)} - 1 \simeq e^{h\nu/(k_B T)}$$

and hence

$$u_P(\nu, T) \simeq \frac{8\pi h\nu^3}{c^3} e^{-h\nu/(k_B T)} \rightarrow 0.$$

The exponential suppression overcomes the factor ν^3 , giving a finite high-frequency tail and a finite total energy.

- (d) Exciting a mode by one quantum requires the energy $h\nu$. At fixed temperature, a high-frequency mode has $h\nu \gg k_B T$, so thermal fluctuations rarely provide enough energy to excite it. Its occupation is therefore strongly suppressed. Classical equipartition assigns an average energy of order $k_B T$ to every mode, regardless of frequency, and misses this suppression.
- (e) From $h\nu_{\max}/(k_B T) = 2.821$,

$$\begin{aligned}\nu_{\max} &= 2.821 \frac{k_B T}{h} \\ &= 2.821 \frac{(1.381 \times 10^{-23} \text{ J/K})(5800 \text{ K})}{6.626 \times 10^{-34} \text{ J s}} \\ &\simeq \boxed{3.41 \times 10^{14} \text{ Hz}}.\end{aligned}$$

Since $\nu_{\max} \propto T$, doubling the temperature doubles $\nu_{\max}$.

Problem 3. The Photoelectric Effect (20 points)

- (a) At threshold, the photon energy is just equal to the work function:

$$h\nu_0 = W.$$

Thus

$$\nu_0 = \frac{W}{h} = \frac{2.30 \text{ eV}}{4.136 \times 10^{-15} \text{ eV s}} = \boxed{5.56 \times 10^{14} \text{ Hz}}.$$

- (b) The photon energy is

$$E_\gamma = h\nu = (4.136 \times 10^{-15} \text{ eV s})(8.00 \times 10^{14} \text{ s}^{-1}) = \boxed{3.31 \text{ eV}}.$$

Therefore

$$K_{\max} = E_\gamma - W = 3.3088 \text{ eV} - 2.30 \text{ eV} = \boxed{1.01 \text{ eV}}.$$

- (c) Because $1 \text{ eV} = q_e(1 \text{ V})$, the numerical value of the stopping voltage in volts equals the maximum kinetic energy in electron-volts:

$$|V_0| = \frac{K_{\max}}{q_e} = \boxed{1.01 \text{ V}}.$$

- (d) At fixed frequency, doubling the intensity doubles the number of photons arriving per unit time. Ideally, this doubles the photocurrent. The energy of each photon, $h\nu$, is unchanged, so both $K_{\max}$ and $|V_0|$ remain unchanged.

(e) For the new source,

$$E_\gamma = h\nu = (4.136 \times 10^{-15})(4.00 \times 10^{14}) \text{ eV} = 1.65 \text{ eV}.$$

Since $1.65 \text{ eV} < W = 2.30 \text{ eV}$, each photon is short by about 0.646 eV . Thus no electrons are emitted in the one-photon process, no matter how large the intensity is.

Problem 4. Young's Double-Slit Experiment (20 points)

(a) Constructive interference occurs when the path difference is an integer number of wavelengths:

$$\Delta r = \ell \sin \theta = n\lambda, \quad n \in \mathbb{Z}.$$

For small angles, $\sin \theta \simeq \tan \theta \simeq y/D$, so

$$\ell \frac{y_n}{D} \simeq n\lambda \quad \Longrightarrow \quad \boxed{y_n \simeq \frac{D\lambda}{\ell} n}.$$

(b) The fringe spacing is

$$\begin{aligned} \Delta y &= \frac{D\lambda}{\ell} \\ &= \frac{(2.00 \text{ m})(500 \times 10^{-9} \text{ m})}{0.250 \times 10^{-3} \text{ m}} \\ &= 4.00 \times 10^{-3} \text{ m} = \boxed{4.00 \text{ mm}}. \end{aligned}$$

Therefore

$$\boxed{y_0 = 0, \quad y_{\pm 1} = \pm 4.00 \text{ mm}, \quad y_{\pm 2} = \pm 8.00 \text{ mm}.}$$

(c) Let $\Delta\theta = \theta_1 - \theta_2$. Then

$$\begin{aligned} |A_1 + A_2|^2 &= (A_1 + A_2)^*(A_1 + A_2) \\ &= |A_1|^2 + |A_2|^2 + A_1^* A_2 + A_2^* A_1 \\ &= 2A_0^2 + A_0^2 e^{-i\Delta\theta} + A_0^2 e^{i\Delta\theta} \\ &= \boxed{2A_0^2 [1 + \cos(\Delta\theta)]}. \end{aligned}$$

The cross term $A_1^* A_2 + A_2^* A_1 = 2A_0^2 \cos(\Delta\theta)$ is responsible for interference. Adding the two one-slit intensities omits this term and therefore cannot produce destructive-interference minima.

- (d) At $y = 0$, the phase is zero:

$$I(0) = 2I_0(1 + \cos 0) = \boxed{4I_0},$$

so the center is bright. Since $\Delta y = D\lambda/\ell$, at $y = \Delta y/2$ the phase is

$$\frac{2\pi\ell}{\lambda D} \frac{\Delta y}{2} = \pi.$$

Therefore

$$I(\Delta y/2) = 2I_0(1 + \cos \pi) = \boxed{0},$$

so this point is dark.

- (e) One electron produces one localized detection on the screen. After many electrons, the accumulated detections form an interference pattern. If the apparatus leaves reliable which-slit information, the paths become distinguishable and the interference pattern disappears.

Problem 5. Deriving the de Broglie Relation (20 points)

- (a) For a photon,

$$p = \frac{E}{c} = \frac{h\nu}{c}.$$

Since $c = \lambda\nu$, we have $\nu/c = 1/\lambda$, and therefore

$$\boxed{p = \frac{h}{\lambda}}.$$

Also, $1/\lambda = k/(2\pi)$, so

$$p = \frac{h}{\lambda} = \frac{h}{2\pi} k = \boxed{\hbar k}.$$

- (b) De Broglie hypothesized that the same wavelength–momentum relation applies to matter:

$$\boxed{\lambda = \frac{h}{p}, \quad p = \hbar k.}$$

Part (a) derives the relation for photons using relations already known for light. Applying it to a massive particle is an additional physical hypothesis, later confirmed by electron diffraction.

- (c) Conservation of energy gives

$$q_e V_{\text{acc}} = \frac{p^2}{2m_e} \implies p = \sqrt{2m_e q_e V_{\text{acc}}}.$$

Hence

$$\lambda = \frac{h}{p} = \boxed{\frac{h}{\sqrt{2m_e q_e V_{\text{acc}}}}}.$$

Numerically,

$$\begin{aligned} p &= \sqrt{2(9.109 \times 10^{-31} \text{ kg})(1.602 \times 10^{-19} \text{ C})(150 \text{ V})} \\ &\simeq \boxed{6.62 \times 10^{-24} \text{ kg m/s}}, \end{aligned}$$

and

$$\lambda = \frac{6.626 \times 10^{-34}}{6.616 \times 10^{-24}} \text{ m} \simeq \boxed{1.00 \times 10^{-10} \text{ m} = 0.100 \text{ nm}}.$$

(d) The wave number is

$$k = \frac{2\pi}{\lambda} = \frac{p}{\hbar} \simeq \boxed{6.27 \times 10^{10} \text{ m}^{-1}}.$$

A plane wave traveling in the positive x direction is

$$\boxed{\psi(x, t) = Ae^{i(kx - \omega t)}}.$$

(e) Since

$$p \propto \sqrt{V_{\text{acc}}}, \quad \lambda \propto \frac{1}{\sqrt{V_{\text{acc}}}}, \quad k \propto \sqrt{V_{\text{acc}}},$$

replacing V_{acc} by $4V_{\text{acc}}$ gives

$$\boxed{p \longrightarrow 2p, \quad \lambda \longrightarrow \frac{\lambda}{2}, \quad k \longrightarrow 2k.}$$

Electron diffraction produces maxima and minima at angles determined by a wavelength. The wavelength inferred from those angles agrees with h/p and changes with accelerating voltage as predicted, providing direct evidence for de Broglie's hypothesis.

End of Solutions